

1. Calculus Language: Foundations

The vocabulary of calculus — algebra, functions, and trigonometry. This refresher covers the essential patterns you'll use constantly.

1.1. 1. Algebra: The Manipulation Rules

Calculus is 90% algebra in disguise. The derivative and integral rules are straightforward; the hard part is simplifying the result. Master these patterns.

1.1.1. Exponents and Logarithms

Exponential Laws:

- Product: $a^m \cdot a^n = a^{m+n}$ — adding exponents when multiplying same bases
- Power of power: $(a^m)^n = a^{mn}$ — multiply exponents when raising powers
- Negative: $a^{-n} = \frac{1}{a^n}$ — flip to denominator
- Fractional: $a^{\frac{m}{n}} = \sqrt[n]{a^m}$ — denominator is the root, numerator is the power

Logarithm Laws:

- Product: $\log(ab) = \log a + \log b$ — logs turn multiplication into addition
- Quotient: $\log\left(\frac{a}{b}\right) = \log a - \log b$ — division becomes subtraction
- Power: $\log(a^b) = b \log a$ — exponents come out front
- Change of base: $\log_a x = \frac{\log_b x}{\log_b a}$ — crucial for calculators

The natural logarithm $\ln x = \log_e x$ uses $e \approx 2.71828$. In calculus, e is special because $(e^x)' = e^x$ — the function is its own derivative.

1.1.2. Factoring Patterns

Recognize these instantly:

- **Difference of squares:** $a^2 - b^2 = (a - b)(a + b)$
- **Difference of cubes:** $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- **Sum of cubes:** $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- **Perfect square:** $a^2 + 2ab + b^2 = (a + b)^2$

The quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ solves $ax^2 + bx + c = 0$. The discriminant $\Delta = b^2 - 4ac$ tells you: positive $\rightarrow$ two real roots, zero $\rightarrow$ one repeated root, negative $\rightarrow$ complex roots.

1.1.3. Summation and Series

The geometric series appears everywhere in finance and analysis:

$$\sum_{i=0}^n r^i = \frac{1 - r^{n+1}}{1 - r}$$

for $r \neq 1$

For $|r| < 1$, as $n \rightarrow \infty$: $\sum_{i=0}^{\infty} r^i = \frac{1}{1-r}$

1.2. 2. Functions: Mappings and Behavior

A function $f : A \rightarrow B$ assigns exactly one output to each input. In calculus, we care deeply about:

- **Domain:** valid inputs (where is the function defined?)
- **Range:** possible outputs (what values can it produce?)

1.2.1. Function Operations

Composition: $(f \circ g)(x) = f(g(x))$ — apply g first, then f . Order matters: $f(g(x)) \neq g(f(x))$ in general.

Inverses: $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$. An inverse exists only if f is **injective** (one-to-one): each output comes from exactly one input. The horizontal line test checks this visually.

1.2.2. Transformations (The “Inside vs Outside” Rule)

Given $y = f(x)$:

Outside changes (applied to output $f(x)$):

- $f(x) + c$: shift **up** by c
- $f(x) - c$: shift **down** by c
- $cf(x)$: vertical stretch by c (or compression if $0 < c < 1$)
- $-f(x)$: reflect over **x-axis**

Inside changes (applied to input x):

- $f(x + c)$: shift **left** by c (counterintuitive!)
- $f(x - c)$: shift **right** by c
- $f(cx)$: horizontal **compression** by c (if $c > 1$)
- $f(-x)$: reflect over **y-axis**

Memory aid: “Inside changes behave opposite to intuition.” Adding inside shifts left; multiplying inside compresses horizontally.

1.2.3. Function Families

Polynomials: $f(x) = a_n x^n + \dots + a_1 x + a_0$

- Degree n determines end behavior
- Leading coefficient a_n determines direction (positive $\rightarrow$ right end goes to $+\infty$ for even degree)

Exponentials: $f(x) = a^x$ with $a > 0, a \neq 1$

- Always positive
- Pass through $(0, 1)$ since $a^0 = 1$
- Growth if $a > 1$, decay if $0 < a < 1$

Logarithms: $f(x) = \log_a x$

- Domain: $x > 0$ only
- Pass through $(1, 0)$ since $\log_a 1 = 0$
- Inverse of exponentials: $\log_{a(a^x)} = x$

1.3. 3. Trigonometry: The Circular Functions

Calculus lives and breathes sine and cosine. You must know their derivatives, integrals, and identities cold.

1.3.1. The Pythagorean Identity

$$\sin^2 x + \cos^2 x = 1$$

This is the foundation. Divide by $\cos^2 x$ to get: $1 + \tan^2 x = \sec^2 x$ Divide by $\sin^2 x$ to get: $1 + \cot^2 x = \csc^2 x$

1.3.2. Angle Sum and Difference

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$

$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$$

Notice the pattern: sine keeps the same sign, cosine flips it. These are crucial for integration techniques later.

1.3.3. Double Angle Formulas

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

The multiple forms of $\cos(2x)$ are useful for different situations — learn to recognize when each applies.

1.3.4. Half-Angle Formulas

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

These are **power-reduction** formulas. They convert squared trig functions into linear ones — essential for integrating $\sin^2 x$ or $\cos^2 x$.

1.3.5. Common Values (Know Cold)

Degrees	Radians	sin	cos	tan
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	undefined

1.4. 4. Domain Restrictions in Calculus

These restrictions appear constantly:

- $\frac{1}{x}$: undefined at $x = 0$ (vertical asymptote)
- $\sqrt{x}$: requires $x \geq 0$ (domain restriction)
- $\ln x$: requires $x > 0$ (asymptotic at 0)
- $\tan x$: undefined at $x = \frac{\pi}{2} + n\pi$ (vertical asymptotes)
- $\arcsin x, \arccos x$: domain $[-1, 1]$ only

1.5. Reflection Questions

1. **Why is e special in calculus?** (Hint: think about derivatives)
2. **If you compose $f(x) = x^2$ and $g(x) = x + 1$, what's $f(g(x))$ vs $g(f(x))$?** Calculate both and compare.
3. **Explain in your own words:** Why does $f(x + c)$ shift left instead of right?
4. **Derive the identity:** $\tan^2 x + 1 = \sec^2 x$ starting from $\sin^2 x + \cos^2 x = 1$.
5. **When would you use the power-reduction formulas?** Give an example where they're necessary.

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